

Design and Bio-Mimetic Analysis of a 2-DOF EMG Controlled Prosthetic Wrist Joint Using Additive Manufacturing

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Research Area (Short Introduction)

EMG-Controlled Prosthetic Wrist Technology

This research focuses on developing **advanced prosthetic wrist joints** that closely mimic natural human wrist movements using electromyography (EMG) control systems and additive manufacturing.

Key Components:

- **Two Degrees of Freedom (2-DOF):** Flexion-extension and supination-pronation movements
- **EMG Control:** Muscle signal-based intuitive control system
- **3D Printing/Additive Manufacturing:** Cost-effective fabrication using PLA material
- **Bio-mimetic Design:** Human-like motion replication with specialized gear systems

The human wrist supports multiple complex motions including flexion (70-90°), extension (65-85°), supination (80-90°), and pronation (75-90°), which are essential for daily activities.

Gaps in the Area

Current Limitations in Prosthetic Wrist Technology

Major Technological Gaps:

① Limited Motion Capability

- Most commercial prosthetic wrists restricted to single motion
- Lack of multi-DOF functionality leads to unnatural movements

② Manufacturing and Cost Challenges

- High costs of existing multi-DOF designs
- Complex manufacturing processes and limited customization
- Heavy and bulky designs affecting user comfort

③ Control System Limitations

- Single-channel EMG systems limiting simultaneous motion control
- High response latency (greater than 250 ms) in existing systems

4 Material and Fabrication Issues

- Limited use of additive manufacturing for customization
- Durability concerns with biocompatible materials
- Insufficient integration of lightweight designs

5 Validation Deficiencies

- Lack of rigorous bio-mimetic validation against human wrist motion
- Limited quantitative analysis of kinematic performance
- Insufficient testing in real-world daily activities

Research Problem

Problem Statement

Primary Research Challenge:

How to develop a cost-effective, lightweight, and biomimetically accurate 2-DOF prosthetic wrist joint that provides natural human-like motion while maintaining intuitive EMG-based control?

Specific Problems Addressed:

- 1 **Functional Limitations:** Current prosthetic wrists cannot replicate natural flexion-extension and supination-pronation movements simultaneously
- 2 **Accessibility and Cost:** High manufacturing costs make advanced prosthetic wrists inaccessible to many users
- 3 **Bio-mechanical Accuracy:** Existing designs fail to achieve human-like kinematics with sufficient correlation
- 4 **Control Integration:** Need for intuitive control that responds naturally to user's muscle signals

How to Solve the Problem

Proposed Solution Methodology

1. Mechanical Design Approach

- **Two-DOF Kinematic System:** Parallel and serial kinematic principles
- **Specialized Gear System:** Semi-spherical palm gear with curved-tooth monopole gear
- **Roller Hub Mechanism:** Low-friction bearing enclosure for smooth rotation
- **Compact Integration:** Design optimized for prosthetic hand integration

2. Additive Manufacturing Strategy

- **Material:** Polylactic Acid (PLA) - biocompatible, cost-effective
- **3D Printing Parameters:** Layer height: 0.2 mm, Infill: 20 percent, Temp: 200°C
- **Design Optimization:** CAD models optimized for FDM printing

How to Solve the Problem (Continued)

3. EMG Control System Implementation

- **Hardware:** Arduino Uno, AD8232 ECG sensor, SG90 servo motors
- **Signal Processing:** Threshold-based control logic (0.1-1.5 mV)
- **Calibration:** EMG signal mapping to servo positions

4. Bio-mimetic Design Principles

- **Range of Motion:** $\pm 90^\circ$ supination-pronation, $\pm 80^\circ$ flexion-extension
- **Force Specifications:** Design for typical daily activity loads (80-120 N)
- **Weight Optimization:** Target lightweight design (approximately 250g)
- **Kinematic Accuracy:** Specialized gear geometry for human-like motion

How to Validate the Solution

Validation and Testing Methodology

1. Kinematic Performance Analysis

- **Motion Tracking:** Kinovea software for joint angle tracking
- **Comparative Study:** Side-by-side comparison with human wrist movements
- **Video Analysis:** 30 fps recording with marker placement
- **Synchronization:** Dynamic Time Warping (DTW) for speed variations

2. Quantitative Metrics

- **Mean Absolute Error (MAE):** Average deviation from human motion
- **Root Mean Square Error (RMSE):** Overall trajectory accuracy
- **Pearson's Correlation:** Similarity between human and prosthetic motion
- **Target Performance:** Correlation coefficients greater than 95 percent

Validation Results

3. Biomechanical Validation Results

Motion	Correlation (r)	MAE	RMSE
Supination	0.9510	0.1796	0.2040
Pronation	0.9848	0.1048	0.1268
Flexion	0.9906	0.0441	0.0651
Extension	0.9787	0.1163	0.1412

4. Additional Validation

- **Mechanical:** Range of motion testing, durability testing (greater than 100 cycles)
- **Control System:** EMG signal calibration, response time (approximately 250 ms)
- **Manufacturing Quality:** Dimensional accuracy, material performance

Validation Outcome: Strong bio-mimetic performance with correlation coefficients above 95 percent for all movements.

Thank You!

Questions and Discussion